Ishan Sinha

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ishansinha5.github.io/ishan-sinha-portfolio/

Education

Michigan State University East Lansing, MI
Bachelor of Science in Computer Science | Concentration: Artificial Intelligence *Aug. 2024 – Dec. 2027*
- Minors: Data Science; Computational Math, Science & Engineering; Mathematics | **Michigander Scholar**

Experience

Auto-Owners Insurance Lansing, MI
Software Development Intern *May 2026 – Aug. 2026*
- Orchestrating enterprise configuration updates within Guidewire Claim Center to ensure high availability for large-scale claims.
- Optimizing backend software configurations for production environments, reducing latency in claims processing workflows.

Facility for Rare Isotope Beams (FRIB) East Lansing, MI
Undergraduate Researcher *Jan. 2026 – Present*
- Engineered a 3D sparse neural network framework, accelerating rare event detection in high-dimensional physical datasets.
- Optimized sparse tensor operations to maximize compute efficiency, significantly reducing latency for 3D structural analysis.

MSU Unmanned Systems East Lansing, MI
Software Team Lead *Nov. 2025 – Present*
- Directed software lifecycle for autonomous drone missions, deploying proprietary ML models for aerial object detection.
- Programmed autonomous flight paths using ArduPilot telemetry analysis, ensuring precision and mission compliance.

Encora Remote
Software Development Intern *Jul. 2023 – Aug. 2023*
- Developed a scalable polling web application using Django, focusing on full-stack integration and clean API documentation.
- Engineered a real-time data processing module to rank and filter user-generated poll data with sub-second response times.

Projects

NBA Finals NLP Predictor | *Python, PyTorch, Whisper, ChromaDB, Streamlit*
- Synthesized emotional feature vectors via RoBERTa, leveraging Whisper for high-fidelity transcription of noisy media.
- Implemented a RAG pipeline utilizing ChromaDB and semantic chunking to ground predictive analysis across 1000+ empirical transcript snippets.
- Architected for local inference to bypass cloud-compute dependency, delivering low-carbon, high-performance analytics.

Comic Book Shop Detector & Recommender | *Python, Java (Spring Boot), C++, PostgreSQL*
- Refactored web app into a high-performance microservices pipeline to handle complex spatial data and inventory.
- Architected a C++ perceptual hashing (pHash) layer to intercept duplicate images, eliminating redundant GPU cycles.
- Containerized a modular Docker-based architecture separating the Java backend, Python vision module, and C++ service.

Technical Skills

Languages: Python, Java, C/C++, C#, JavaScript, SQL, HTML/CSS
Frameworks: PyTorch, Scikit-Learn, Hugging Face (RoBERTa, Whisper), Spring Boot, FastAPI, ChromaDB
Proficiencies: RAG, Microservices Architecture, Compute Optimization, Unsupervised Learning, Autonomous Systems
Tools: Git, GitHub, GitHub Actions, Docker, CI/CD Pipelines, REST APIs, Linux/Unix